

Effect of vitamin C supplements on physical performance.



Vitamin C is an essential component of the diet and may reduce the adverse effects of exercise-induced reactive oxygen species, including muscle damage, immune dysfunction, and fatigue. However, reactive oxygen species may mediate beneficial training adaptations that vitamin C attenuates; indeed, from a total of 12 studies, vitamin C in doses $>1 \text{ g}\cdot\text{d}^{-1}$ impaired sport performance substantially in four of four studies, possibly by reducing mitochondrial biogenesis, while a further four studies demonstrated impairments that were not statistically significant. Doses of $\sim 0.2 \text{ g}\cdot\text{d}^{-1}$ of vitamin C consumed through five or more servings of fruit and vegetables may be sufficient to reduce oxidative stress and provide other health benefits without impairing training adaptations.